Joint Inventory and Routing Optimization for Multi-City Logistics: A Case Study on Meituan Fresh

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Abstract

Abstract— The "last-mile" fresh-food delivery network faces a fundamental trade-off between inventory holding costs and transportation costs. In this work, we propose a joint optimization framework for a five-city, 25-station logistics network. For each city, we optimize the replenishment cycle T by balancing storage and routing costs; subsequently, we enumerate all $7^5 = 16{,}807$ global T-combinations to identify the system-wide optimum. The best configuration achieves a total daily cost of **24,874 CNY**, with city-level optimal cycles ranging from $T^* = 1$ to $T^* = 6$. Compared with uniform-T baselines, the optimal differentiated strategy saves up to **20,058 CNY/day** (80.6%). Our results demonstrate that city-specific replenishment policies are essential for cost-efficient operations in heterogeneous logistics networks.

1 Introduction

Fresh e-commerce platforms such as Meituan face the daily challenge of replenishing perishable goods from a central distribution center (CDC) to multiple city hubs and then to retail stations. A longer replenishment cycle T reduces the frequency of long-haul transportation but inflates average inventory levels and holding costs; conversely, a shorter T cuts storage expenses at the expense of higher transport spending. Existing practice often treats inventory and routing decisions separately, or imposes a uniform T across all cities, which frequently leads to sub-optimal outcomes.

This paper addresses the following two questions for a real-world Meituan network:

1. *City-level*: What is the optimal replenishment cycle T^* for each city, and how do storage and transport costs interact?
2. *System-level*: How much additional value can a *global joint optimization* of all cities' T provide over independent optimization and uniform-T policies?

2 Problem Statement

The network consists of one CDC and five city hubs (A–E). Each hub serves a set of retail stations with known daily demand and distances to the CDC. Two vehicle types are available: large trucks (capacity 200,000 boxes) and small trucks (capacity 80,000 boxes). The planning objective is to minimize the *average daily total cost*, which comprises (i) inventory holding cost at the city hub and (ii) daily transportation cost, including both CDC-to-city long-haul and city-to-station local delivery.

The decision variables are the integer replenishment cycles $T_i \in \{1, \ldots, 7\}$ for each city $i \in \{A, \ldots, E\}$, together with the vehicle routing plans that satisfy capacity constraints in each replenishment trip.

3 Methodology

3.1 City-Level Inventory–Transport Model

For a given city and a candidate cycle T, the average daily cost is

$$C(T) = C_s(T) + C_t(T), \tag{1}$$

where the storage cost is proportional to the cycle length and the daily demand d:

$$C_s(T) = h \cdot \frac{T \cdot d}{2}, \tag{2}$$

with h the unit holding cost. The transport cost contains two parts: the CDC-to-city long-haul cost (a fixed cost per trip divided by T) and the city-to-station local routing cost. The local routing problem is solved by exhaustive grouping: we enumerate all feasible partitions of stations into groups of size at most three, compute the cheapest eligible vehicle mix for each group, and pick the partition with the minimum single-trip cost. The resulting single-trip transport cost is amortized over T days:

$$C_t(T) = \frac{1}{T}\Big(f_{\text{CDC}} + \sum_{g \in \mathcal{G}} c(g)\Big), \tag{3}$$

where $\mathcal{G}$ is the set of station groups and $c(g)$ the routing cost for group g under its assigned vehicle. City-level optimization simply enumerates $T = 1, \ldots, 7$ and picks the minimizer T^*.

3.2 Vehicle Routing and Load-Rate Monitoring

For every station group, the model selects the cheapest feasible vehicle mix (small and/or large trucks) and constructs a detailed unloading sequence. Because vehicles are filled in station order, the model records the load rate both *before* and *after* each stop. This load-rate trace serves two purposes: (i) it verifies that capacity constraints are respected throughout the tour, and (ii) it highlights under-utilized trips. For example, City B's group $[B_4, B_5]$ is served by a large truck at only 42.5% load, whereas City A's group $[A_3, A_4, A_5]$ reaches 95%. Such disparities motivate dynamic re-grouping or a switch to smaller vehicles when demand patterns shift.

3.3 Global Joint Optimization

Because there are only five cities and $T \in \{1, \ldots, 7\}$, the full decision space contains $7^5 = 16{,}807$ combinations. Global optimality is guaranteed by exhaustive enumeration: for each combination $(T_A, \ldots, T_E)$, we sum the corresponding city-level optimal costs and select the combination with the minimum total cost.

Algorithm 1 City-Level Inventory Optimization

Require: City code c
for $T = 1$ to 7 **do**
  $S \leftarrow$ compute_storage_cost(c, T)
  $G \leftarrow$ exhaustive_station_grouping(c, T)
  $R \leftarrow$ assign_vehicles(G)
  $V \leftarrow$ compute_trip_cost$(R)/T$
  cost$[T] \leftarrow S + V$
end for
return $T^* = \arg\min_T \text{cost}[T]$, cost$[T^*]$

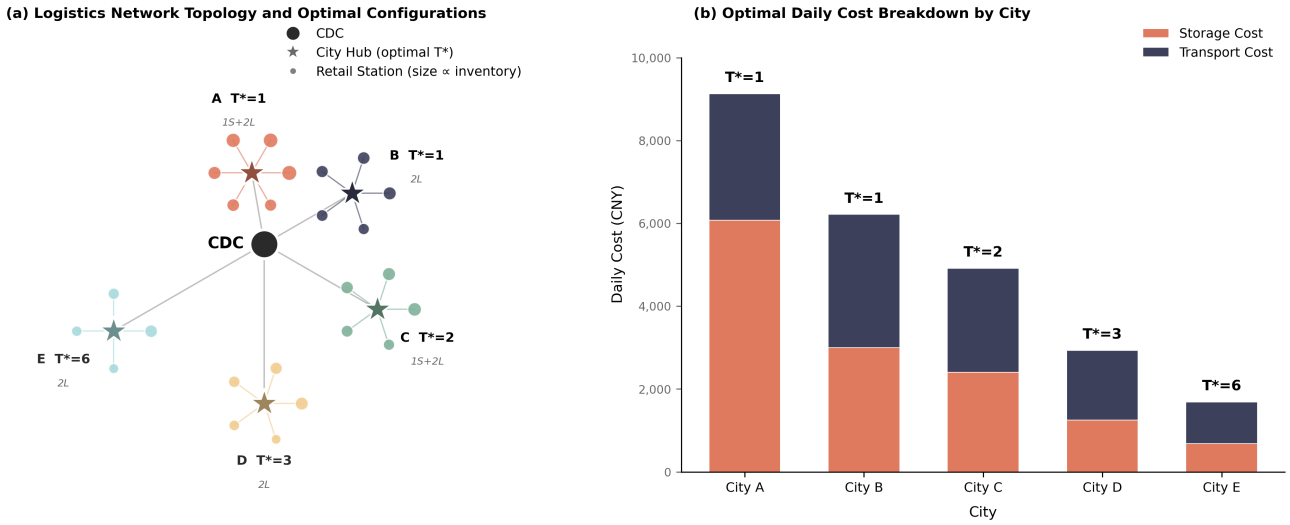

Figure 1: (a) Logistics network topology: one CDC serves five city hubs (A–E). Each hub is annotated with its optimal replenishment cycle T^* and the required fleet size. (b) Optimal daily cost breakdown by city, split into storage cost (orange) and transport cost (navy).

4 City-Level Results and Sensitivity

4.1 Sensitivity Analysis

Figure 2 plots the storage, transport, and total costs for each city as T varies from 1 to 7 days.

Cities A and B exhibit monotonically increasing total costs. Because both cities have large daily demand, the storage cost $C_s(T)$ grows almost linearly with T and quickly dominates the modest savings in transport. Consequently, the optimal cycles are boundary solutions at $T^* = 1$.

Table 1: Optimal configurations and cost breakdown for the five cities.

City	T^*	Stations	Trucks	Storage (CNY)	Transport (CNY)	Total (CNY)
A	1	6	3	6,075.0	3,055.0	9,130.0
B	1	5	2	3,000.0	3,215.0	6,215.0
C	2	5	3	2,400.0	2,515.0	4,915.0
D	3	5	2	1,250.0	1,678.3	2,928.3
E	6	4	2	684.0	1,001.7	1,685.7

City C shows a weak U-shape with the minimum at $T^* = 2$. At $T = 1$ the transport cost is still high; extending the cycle to two days yields just enough transport savings to outweigh the additional inventory cost.

City D presents a classic U-shaped curve with a clear interior optimum at $T^* = 3$. For $T < 3$ the transport penalty is severe; for $T > 3$ the holding cost rebounds and overtakes the diminishing transport savings.

City E, having the smallest demand, can tolerate the longest cycle. The total cost declines until $T^* = 6$, after which the inventory increase at $T = 7$ pushes the cost upward again.

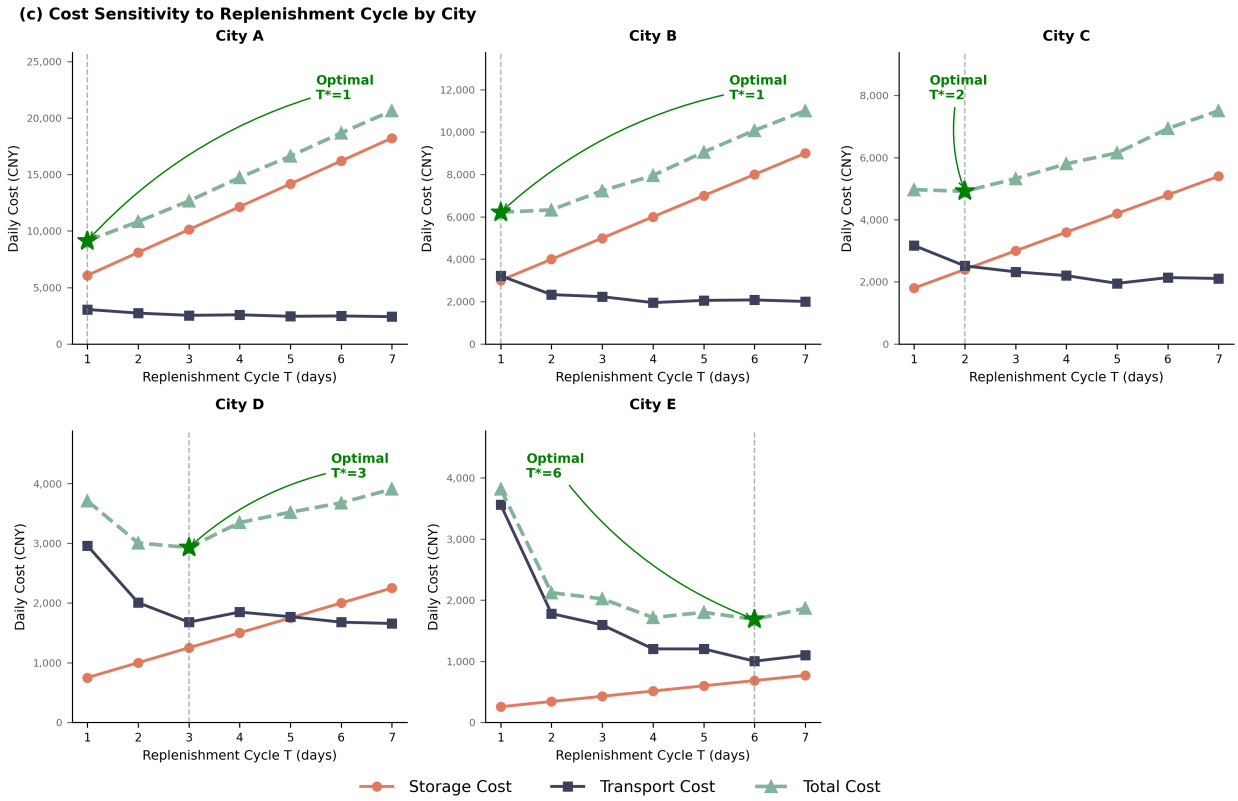

Figure 2: Cost sensitivity to replenishment cycle T for cities A–E. The green star marks the optimal T^* for each city.

5 Global Optimization and Managerial Insights

5.1 System-Wide Performance

The global optimum coincides with the independent-optimization sum: $T^* = \{A{:}1, B{:}1, C{:}2, D{:}3, E{:}6\}$ yields a total daily cost of **24,874 CNY**. Although there is no coordination penalty in this particular instance, the global enumeration reveals that the solution landscape is steep: the top-10 alternatives are only 30–145 CNY/day more expensive (Figure 3, left), confirming the robustness of the optimal configuration.

More importantly, the value of *differentiation* is stark when compared against uniform-T policies (Figure 3, right). Forcing all cities to $T = 1$ raises the daily cost by 2,968 CNY (+11.9%); a uniform $T = 3$ adds 5,290 CNY (+21.3%); and a uniform $T = 7$ explodes the cost by 20,058 CNY (+80.6%). These gaps underscore that a one-size-fits-all policy is economically unacceptable in a heterogeneous network.

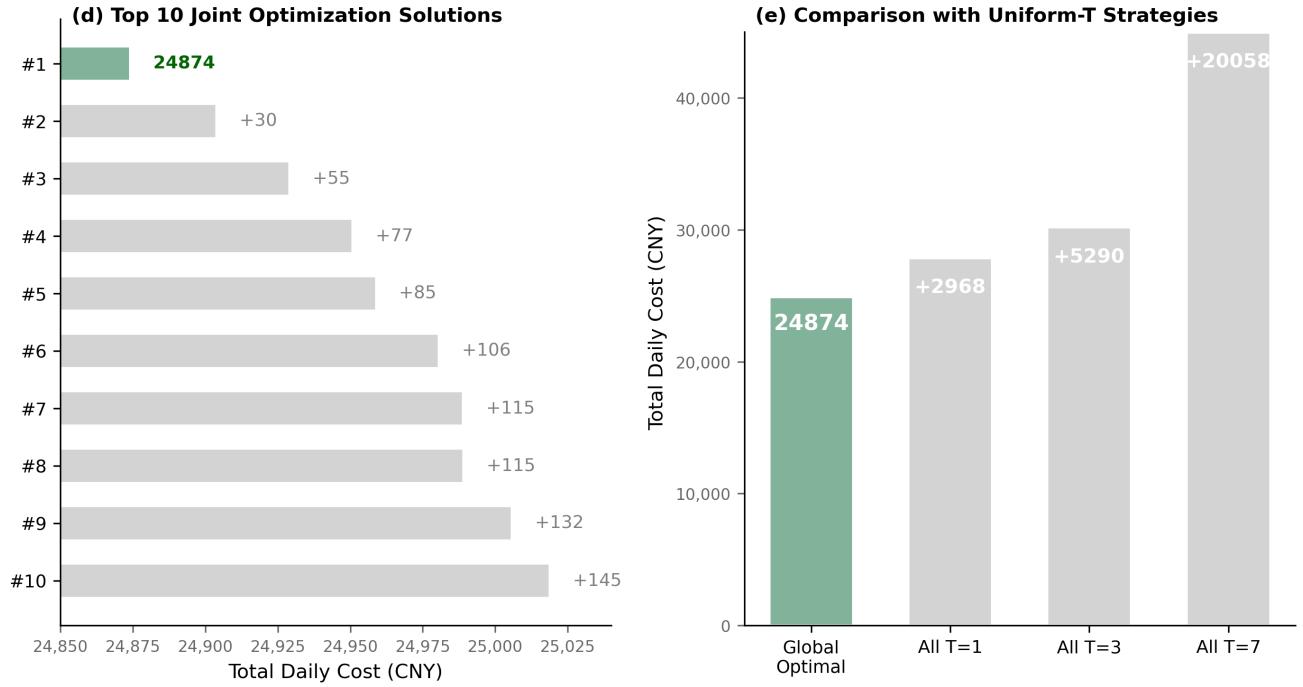

Figure 3: (Left) Top 10 joint-optimization solutions ranked by total daily cost; the optimal solution is highlighted in green. (Right) Comparison of the global optimal strategy with uniform-T baselines; the numbers above the bars show the incremental daily cost relative to the optimum.

5.2 Managerial Insights

We distill three actionable recommendations for operations managers:

1. **Reject uniform policies.** Cities with high demand (A, B) must be replenished daily, whereas low-demand cities (E) can operate on much longer cycles. Adopting a differentiated T policy is the single largest lever for cost reduction.

2. **Watch the storage–transport inflection point.** For medium-demand cities such as D, the total-cost curve is U-shaped. Precise calibration of the inflection point ($T^* = 3$ in this case) can yield savings of several hundred CNY per day.

3. **Coordinate fleet planning with T.** Extending T increases single-trip volume. Managers should simultaneously review whether larger vehicles or revised station groupings are needed to keep local routing efficient.

4. **Monitor vehicle load rates.** Low load rates signal idle capacity and inflated per-unit transport costs. For under-utilized groups (e.g., 42.5% for $[B_4, B_5]$), managers should consider merging additional nearby stations or switching to smaller trucks to improve asset utilization.

6 Conclusion

This study proposes a computationally tractable joint inventory–routing framework for a multi-city fresh-food logistics network. By combining city-level T-optimization with global enumeration, we identify a differentiated replenishment strategy that achieves a minimum daily cost of 24,874 CNY and dramatically outperforms uniform-T baselines. Future work will extend the model to stochastic demand and dynamic T adjustment to further enhance operational resilience.